

Neoemantix - Assessment

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Question 1:

A) What is “spaCy”?

Answer:

spaCy is an “Open Source” library / framework which could be used for several Natural Language Processing based tasks.

B) What is it used for?

Answer:

spaCy is used for several NLP tasks including:

- POS Tagging
- NER
- Dependency recognition
- Text Classification
- Vectorization and semantic similarity
- Visualization (NER, POS, Dependencies)
- A Complete pipeline for training LLMs
- Flexibility of Models (Can use default CNN based models / Transformer models / Personalized NN models)
- GPU utilization for these tasks for better accuracy and speed.

C) For what functionality of spaCy, GPU is preferred?

Answer:

GPU in spaCy is used for:

- Faster results with existing CNN based models ie words per second
- CPU based model eg: en_core_web_sm | GPU based transformer model: en_core_web_trf
There are examples of CPU based and GPU based pipelines offered by spaCY
- Can also use any other Transformer based model here for NER, POS, Text Classification like BERT based or RoBERTa based.
Eg: if it's a low resource language (not from 75+ claimed language) NER/POS based on these specific languages can be used.
- Since spaCy also provides a pipeline for training of LLMs, GPU is essential here to accelerate the process.

D) Can a regular CPU undertake that functionality?

Answer:

As I mentioned in last answer that CPU optimized pipelines are also present in spaCy like `en_core_web_sm` but in this case we might have to compromise on the accuracy, speed of the desired results.

eg: I watched a video where the problem was to search for Go Programming language in a given set of texts. But one example mentioned "... go programming" and here it considered go as a 'VERB' which was a misclassification. In this case if had used a better model we could have resolved this issue.

E) What options are available in newer versions of "spaCy" in this regard?

Answer:

- **3.7 - Curated Transformers:** This replaces the use of HuggingFace Transformer to connect to API of other LLMs instead this is embedded in the spaCy itself
- **3.6 - Pipeline Update:** Danish and Slovenian pipelines are updated (improve handling of apostrophes and to lemmatize "get" as a passive auxiliary.)
- **3.5 - Better Speed Evaluation:** Here few extra things are considered for accurate evaluation:
 - i) First run on GPU might encounter lag so it runs several times to avoid it
 - ii) shuffles the document for varied workload
 - iii) Better outlier detection basically tells if an outlier in document is creating a lag or its the hardware

F) If you were to measure the performance difference between GPU and CPU for this spaCy functionality, which benchmarking test would you propose?

Answer:

- **Existing Benchmarking in the spaCy documentation:**
 - i) Uses speed of the pipeline in words per second for GPU vs CPU
 - ii) Uses accuracy which tracks metrics like precision, recall, and F-score
- **Proposed Benchmarking Test:**

By using incremental benchmarking, we can determine the break-even point where a GPU begins to outperform a CPU for a specific dataset. This allows us to recommend the most efficient hardware based on the task's batch size and complexity.